

5-Mark Questions & Answers — DBMS Module 3: SQL Standards & Advanced Queries

Q1. Explain Nested Subqueries (IN, NOT IN, EXISTS, NOT EXISTS, ALL, ANY) with SQL examples.

- **IN / NOT IN** : Checks membership in subquery result set.
- **EXISTS / NOT EXISTS** : Evaluates to TRUE if subquery returns at least one row (highly efficient).
- **> ALL / > ANY** : Compares value against all or at least one element of subquery set.
- *Example:* `SELECT Name FROM Emp WHERE Salary > ALL (SELECT Salary FROM Emp WHERE Dept_ID = 20);`

Q2. Explain Recursive Queries and Common Table Expressions (CTE) in SQL.

- **CTE Syntax:** `WITH cte_name AS (query) SELECT * FROM cte_name;`
- **Recursive CTE:** Uses an Anchor Member and a Recursive Member joined by **UNION ALL** to traverse hierarchical structures (organizational charts, graphs).

```
WITH RECURSIVE EmpHierarchy AS (  
    -- Anchor Member (Top Manager)  
    SELECT Emp_ID, Name, Manager_ID, 1 AS Level FROM Employee WHERE Manager_ID IS NULL  
    UNION ALL  
    -- Recursive Member  
    SELECT E.Emp_ID, E.Name, E.Manager_ID, H.Level + 1  
    FROM Employee E INNER JOIN EmpHierarchy H ON E.Manager_ID = H.Emp_ID  
)  
SELECT * FROM EmpHierarchy;
```

Q3. Explain PL/SQL Triggers: Syntax, Timing, Events, and Use Cases.

- Complete syntax for **CREATE OR REPLACE TRIGGER** with **BEFORE/AFTER**, **FOR EACH ROW**, and **:NEW** / **:OLD** qualifiers for audit logging and data validation.